Module 4 - Calibration

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- Topic 1 - Standards and Units
- Topic 2 - The Calibration Curve
- Topic 3 - Linear Regression

Topic 1 - Standards and Units

- Thought Experiment Continued
- Standards and Calibration
- Base Dimensions and Units
- Hierarchy of Standards

Thought Experiment Continued

Thought Experiment: Look around the room and choose an object. It can be anything. Ask yourself the following questions.

- What is the true length of the object?
- What is the physical meaning of that number?
- What do the units relate to?

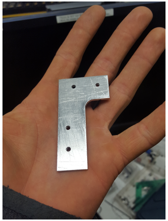


Image: T.Hill

Standards and Calibration

When a measurement system is calibrated, its indicated value is compared directly with a reference value. This reference value forms the basis of the comparison and is known as the standard. This standard may be based on the output from a piece of equipment, from an object having a well- defined physical attribute to be used as a comparison, or from a well-accepted technique known to produce a reliable value.

Text: Theory and Design of Mechanical Measurements, 5th Edition

Standards and Calibration

A dimension defines a physical variable that is used to describe some aspect of a physical system.

A unit defines a quantitative measure of a dimension.

Text: Theory and Design of Mechanical Measurements, 5th Edition

Base Dimensions and Units

The dimension of mass is defined by the kilogram. Originally, the unit of the kilogram was defined by the mass of one liter of water at room temperature. But today an equivalent yet more consistent definition defines the kilogram exactly as the mass of a particular platinum-iridium cylindrical bar that is maintained under very specific conditions at the International Bureau of Weights and Measures located in Sevres, France. This particular bar (consisting of 90% platinum and 10% iridium by mass) forms the primary standard for the kilogram. It remains today as the only basic unit still defined in terms of a material object.

Text: Theory and Design of Mechanical Measurements, 5th Edition

Base Dimensions and Units

The base units relate directly to a standard (historically).

Unit	SI		I-P	
Length	meter	(m)	foot	(ft)
Mass	kilogram	(kg)	slug	$(slug)$
Time	second	(s)	second	(sec)
Temperature	degrees	$(^\circ C, {}^\circ K)$	degrees	$(^\circ F, {}^\circ R)$
Current	ampere	(A)	ampere	(A)
Substance	mole	(mol)	mole	(mol)
Light Intensity	candela	(cd)	candela	(cd)

Base Dimensions and Units

Common derived units are defined in terms of the the base units.

Unit	SI		I-P	
Force	newton	(N)	pound-force	(lb_f)
Voltage	volt	(V)	volt	(V)
Resistance	ohm	(Ω)	ohm	(Ω)
Capacitance	farah	(F)	farah	(F)
Inductance	henry	(H)	henry	(H)
Energy	joule	(J)	foot-pound	$(ft - lb)$
Power	watt	(W)	foot-pound per second	$(ft - lb/sec)$

Hierarchy of Standards

The known value applied to a measurement system during calibration becomes the standard on which the calibration is based.

So how do we pick this standard, and how good is it?

... primary standards are impractical as standards for normal calibration use ... there exists a hierarchy of reference and secondary standards used to duplicate the primary standards.

Primary standard	—	Maintained as absolute unit standard
Transfer standard	—	Used to calibrate local standards
Local standard	—	Used to calibrate working standards
Working standard	—	Used to calibrate local instruments

Text: Theory and Design of Mechanical Measurements, 5th Edition

Hierarchy of Standards

Examples of standards or using a standard:

- -
- -
- -

Topic 2 - The Calibration Curve

- What is Calibration?
- Generalized Curve
- Static Sensitivity and Zero Offet
- Example: IR Distance Sensor

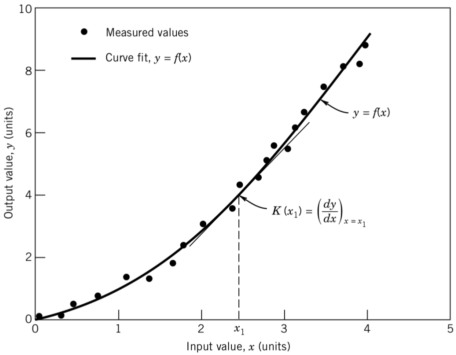
What is Calibration?

A calibration applies a known input value to a measurement system for the purpose of observing the system output value. It establishes the relationship between the input and output values. The known value used for the calibration is called the standard.

- A range of input values can be used to form a calibration curve.
- The calibration curve describes the input-output relationship of the measurement system.

Text: Theory and Design of Mechanical Measurements, 5th Edition,

Static Calibration Curve



y: measured signal (output)

x: known standard (input)

Question: How many values are needed for a calibration? Why?

Image: Theory and Design of Mechanical Measurements, 5th Edition,

Static Sensitivity and Zero Offet

You have learned about these by a a different name.

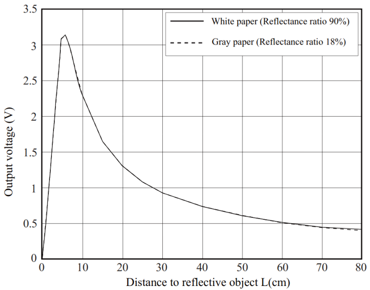
Static Sensitivity - The Slope of the Calibration Curve

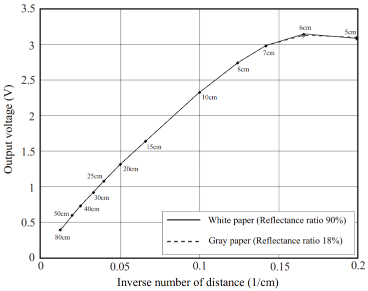
Zero Offset - Y-Intercept of Calibration Curve

Question: How many parameters or variables are needed to describe the curve?

Static Sensitivity and Zero Offet

Example: IR Distance Sensor





Topic 3 - Linear Regression

- Motivation - Functional Relationship
- Least Squares Regression
- Using Software Packages
- Example: IR Sensor Calibration
- Activity: IR Sensor Calibration

Motivation - Functional Relationship

A measured variable is often a function of one or more independent variables that are controlled during the measurement. ... This is a common procedure used to document the relationship between the measured variable and an independent process variable. ...

We can use regression analysis to establish a functional relationship between the dependent variable and the independent variable. This discussion pertains directly to polynomial curve fits.

Other functions such as exponential and log fits can also be used.

Text: T.HIll, Theory and Design of Mechanical Measurements, 5th Edition

Motivation - Functional Relationship

Consider the graphs below. This is a calibration curve.

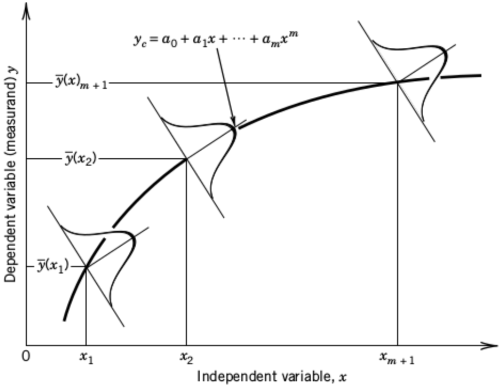


Figure 4.9 Distribution of measured value y about each fixed value of independent variable x. The curve y_c represents a possible functional relationship.

Image:

Theory and Design of Mechanical Measurements, 5th Edition

Least Squares Regression

- We are trying to find a polynomial of best fit for the data.

  $y_c(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_m x^m \qquad m \leq (n-1)$

- This is done by minimizing the quantity below.

  $D = \Sigma_{i=1}^{N}(y_i - y_{ci})^2 = \Sigma_{i=1}^{N}(y_i - a_0 + a_1 x + a_2 x^2 + \cdots + a_m x^m)^2$

- For a first order curve the coefficients become:

  $a_0 = \frac{\Sigma x_i \Sigma x_i y_i - \Sigma x_i^2 \Sigma y_i}{(\Sigma x_i)^2 - N \Sigma x_i^2} \quad , a_1 = \frac{\Sigma x_i \Sigma y_i - N \Sigma x_i y_i}{(\Sigma x_i)^2 - N \Sigma x_i^2}$

Using Software Packages

Most spreadsheet and engineering software packages can perform a regression analysis on a data set. Examples will be shown throughout the course in MATLAB and EXCEL.

- MATLAB - curve fitting toolbox - *polyfit()*
- Excel - *linest()*
- Python - *numpy.linalg.lstsq()*
- Labview - *Linear Fit VI, Exponential Fit VI, etc.*
- and many more...

Sample Calibration Data

Sample #	Known Distance $x_i(m)$	Measured Voltage $y_i(V)$
1	6	1.67
2	9	1.17
3	12	0.89
4	15	0.72
5	18	0.61
6	21	0.53

Linear Least Squares Regression Coefficients: $a_0 =$, $a_1 =$

Functional Relationship:

$y = a_1 * x + a_0$

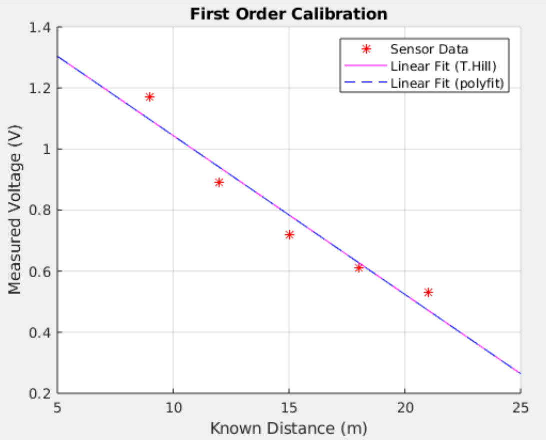
Linear Regression in MATLAB - Manual

```
clear variables;close all;clc

x=[];
y=[];
n=length(x);

% perform Linear Least Squares Regression
a0=
a1=

x_f=5.0:1.0:25;
y_f=

% generate 1st order curve fit with polyfit()
P1=polyfit(x,y,1);

x_f1=5.0:1.0:25;
y_f1=
```

Linear Regression in MATLAB - Manual

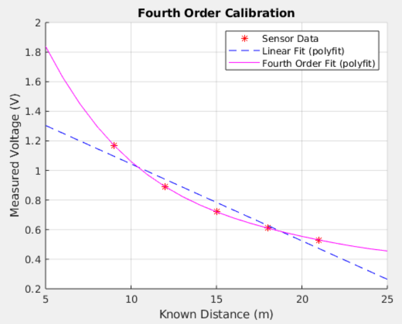
```
figure(1);hold on

plot(x,y,'r*')
plot(x_f,y_f,'m-')
plot(x_f1,y_f1,'b--')

title('First Order Calibration')
xlabel('Known Distance (m)')
ylabel('Measured Voltage (V)')
legend('Raw Data','Linear Fit (T.Hill)','Linear Fit (polyfit)')
grid on
```

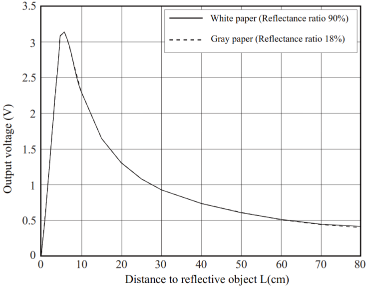
First Order Calibration Curve



Images: T.Hill

Fourth Order Calibration Curve





Images: T.Hill, Sharp

Activity: Sensor Calibration - IR Ranger

Groups of up to 2 - At least one computer is required.

Instructions:

1. Use the sample calibration data provided to calculate the **static sensitivity** and **zero offset** of a 1st order regression line using a MATLAB program.
2. Show the calibration data and calibration curve on the same graph. Label the graph and show a legend.
3. Show how the calibration curve could be used for a measurement application. Show two(2) example data points.
4. Modify your program to calculate an improved calibration curve using a different mathematical model. Discuss the qualities of each curve. Are there benifits and tradeoffs?

Submission:

- Your code should calculate the required values and generate the required graphs.
- Submit your MATLAB code to the appropriate ilearn folder. Include all names on the code.
- Each group member must submit the complete assignment to recieve credit.